

1 Landau-Zener Solution to Kibble-Zurek Scaling

This section comes directly from "Dynamics of a Quantum Phase Transition: Exact Solution in Quantum Ising model"

The time depednent equations for the TFIM reduce to

$$i\hbar \frac{du_k}{d\tau} = -\frac{1}{2}(\tau\Delta_k)u_k + \frac{1}{2}v_k \quad (1)$$

$$i\hbar \frac{dv_k}{d\tau} = +\frac{1}{2}(\tau\Delta_k)v_k + \frac{1}{2}u_k \quad (2)$$

Here, τ depends on the time t and the quench protocol, while Δ_k is the gap parameter for mode k . The functions $u_k(\tau)$ and $v_k(\tau)$ describe the time evolution of the Bogoliubov quasiparticle amplitudes for each momentum mode k .

We can define them as follows:

$$\Delta_k^{-1} = 4J\tau_Q \sin^2(ka) \quad (3)$$

$$\tau = 4J\tau_Q \sin(ka) \left(\frac{t}{\tau_Q} + \cos(ka) \right) \quad (4)$$

This can be solved by first recognizing that only modes iwth small energy contribute in the limit of a large quench:

$$|ka| < \frac{\pi}{4} \quad (5)$$

For these modes we may use the LZ formula for excitation probability:

$$p_k \approx e^{-\frac{\pi}{2\hbar\Delta_k}} \quad (6)$$

This is valid when

$$ka = (4\pi J\tau_Q/\hbar)^{-\frac{1}{2}} << \frac{\pi}{4} \quad (7)$$

or equivalently, when

$$\tau_Q >> \frac{4\hbar}{\pi^3 J} \quad (8)$$

From which we then derive the number of kinks $\mathcal{N}$ or the defect density

$$n = \frac{\mathcal{N}}{N}$$

$$n = \lim_{N \rightarrow \infty} \frac{\mathcal{N}}{N} = \frac{1}{2\pi} \int_{-\pi}^{\pi} d(ka) p_k = \frac{1}{2\pi} \frac{1}{\sqrt{2J\tau_Q/\hbar}} \quad (9)$$

It should be noted that this calculation of defect density provides $O(1)$ corrections to the formula suggested by Zurek. The real highlight is the prediction of power law scaling of defect density and τ_Q :

$$n \propto \tau_Q^{-\frac{1}{2}}$$

The paper also points out that for a finite chain we can also predict the fastest τ_Q that no particles get excited (which can be done because a finite chain is gapped).

This quantity is given as

$$\mathcal{P}_{GS} = \prod_{k>0} (1 - p_k) \quad (10)$$

Only the $\frac{\pi}{N}$ pair is likely to be excited (Question: Why?) so we can approximate that

$$\mathcal{P}_{GS} = 1 - p_{\frac{\pi}{N}} \approx 1 - \exp(-2\pi^3 \frac{J\tau_Q}{\hbar N^2}) \quad (11)$$

So we can estimate that an adiabatic quench occurs when

$$\tau_Q > \tau_Q^{ad} = \frac{\hbar N^2}{2\pi^3 J}$$

Inverting this relationship tells us that the size N of a defect free chain grows as $\tau_Q^{\frac{1}{2}}$

1.1 Ferromagnetic to Paramagnetic

The previous section was for a disordered to ordered quench. Now, we can analyze the opposite.

We can assume the value of g to be of the form

$$g(t > 0) = \frac{t}{\tau_Q}$$

and that when $g = 0$ we are in the even parity groundstate.

In the $g > 1$ groundstate limit we expect all spins to be in the $+x$ orientation, so we may define ground state fidelity as

$$\mathcal{F} = \frac{1}{2} (N - \sum_{n=1}^N \sigma_n^x) \quad (12)$$

when $g \gg 1$ the hamiltonian reduces to

$$H \approx -Jg \sum_{n=1}^N \sigma_n^x = -Jg(N - 2\mathcal{F})$$

At the same time,

$$H^+ \approx 2Jg \sum_k \gamma_k^\dagger \gamma_k - JgN$$

so we can recognize via inspection that

$$\mathcal{F} = \sum_k \gamma_k^\dagger \gamma_k$$

which allows us to predict that defect density is the same,

$$f = \frac{1}{2\pi} \frac{1}{\sqrt{2J\tau_Q/\hbar}} \quad (13)$$

2 Equilibrium Scaling exponents of Projector Operators

For this section, we will slightly modify the hamiltonian by rotation. The underlying physics will remain the same.

$$\hat{H} = -J \sum_{i=1}^L \sigma_i^x \sigma_{i+1}^x - h \sum_{i=1}^L \sigma_i^z \quad (14)$$

Now we will discuss the scaling exponents of the projector operators. We approach this problem in the even parity sector so odd numbers of spin operators give 0 (e.g. $\langle \sigma_x \rangle = 0$)

For starters, we can look at P_2 :

$$\langle P_2 \rangle = \frac{1}{4N} \sum_{i=1}^L (\langle 1 \rangle + \langle \sigma_i^x \sigma_{i+1}^x \rangle) \approx \frac{1}{4} + \frac{\langle \sigma_0^x \sigma_1^x \rangle}{4} \quad (15)$$

and how it is expected to scale near the critical point assuming the thermodynamic limit. Also, assuming PBC then the last approximate equality turns into an exact equality.

2.1 n -point correlation functions

Correlations of the form $\langle \sigma_0^x \sigma_n^x \rangle$ are computed using pfaffians of the fermionic operators. For instance, we can define

$$A_i = c_i^\dagger + c_i \quad (16)$$

and

$$B_i = c_i^\dagger - c_i \quad (17)$$

We can use the definition of σ^x in terms of fermionic creation and annihilation operators to write out the expression

$$\langle \sigma_0^x \sigma_n^x \rangle = \langle \mathbf{B}_0 A_1 B_1 \dots A_{n-1} B_{n-1} \mathbf{A}_n \rangle \quad (18)$$

In the above expression I bold the first and last majorana fermionic operators to emphasize the structure of the correlation functions. The internal structure comes from the nature of the Jordan-Wigner strings that must attach the two operators.

It can be shown through the definition of the operators that

$$\langle A_i A_j \rangle = \delta_{ij} \quad (19)$$

$$\langle B_i B_j \rangle = -\delta_{ij} \quad (20)$$

and that consequently the only "interesting" correlations will be of the form

$$\langle B_i A_j \rangle = -\langle A_j B_i \rangle = D_{i-j+1} \quad (21)$$

There are a few different ways at approaching the calculation of D_n . I cite the solution given by Sachdev:

$$D_n = \int_0^{2\pi} \frac{d\phi}{2\pi} e^{-in\phi} \tilde{D}(e^{i\phi}) \quad (22)$$

$$\tilde{D}(z = e^{i\phi}) = \left(\frac{1-gz}{1-\frac{g}{z}} \right)^{1/2} \tanh \left[\frac{J}{T} \left((1-gz) \left(1 - \frac{g}{z} \right) \right)^{1/2} \right] \quad (23)$$

In the limit where $T = 0$, then the expression for $\tilde{D}$ reduces even more to just

$$\tilde{D} = \left(\frac{1-gz}{1-g/z} \right)^{1/2} \quad (24)$$

The structure above can be used when you are calculating more complicated objects, which I will derive now.

First, let's recognize that we can write

$$\sigma_j^x = (c_j^\dagger + c_j) \prod_{n=0}^{j-1} \sigma_n^z = A_j J(0, j-1) \quad (25)$$

The $J(i, j)$ are the jordan-wigner strings connecting sites i and j . It can be easily seen that they have the property

$$J(i, j) J(i, k) = J(j+1, k) = \prod_{w=j+1}^k i B_w A_w$$

From this, we can see that a general n -point function (assuming n is even) is simple. For example, the 4 point function:

$$\langle \sigma_0^x \sigma_{n_1}^x \sigma_{n_2}^x \sigma_{n_3}^x \rangle = \langle B_0 J(1, n_1 - 1) A_{n_1} B_{n_2} J(n_2 + 1, n_3 - 1) A_{n_3} \rangle \quad (26)$$

The structure of this correlation function generalizes to higher n (assuming n is even). Computationally, n point functions can be made even simpler remembering

$$[\sigma_j^x, \sigma_k^x] = \delta_{j,k}$$

and

$$(\sigma_j^x)^2 = 1$$

Which means we can always rearrange a 4 point function to be arranged from lowest to highest in index, then we can compute correlation functions in the form of connecting two two spins at a time.

If n is odd we need to use cluster decomposition, and we can only recover the magnitude of the correlation function (sign information will be lost). The result generalizes trivially from the 1-point function:

$$|\langle \sigma_j^x \rangle| = \lim_{d \rightarrow \infty} |\langle \sigma_j^x \sigma_{j+d}^x \rangle| \quad (27)$$

In typical canon, the 2-point functions are calculated using Pfaffians of Toeplitz Matrices. The Pfaffian of a matrix $\mathbf{A}$ has the property that

$$\text{PF}(\mathbf{A}) = \sqrt{\det(\mathbf{A})} \quad (28)$$

However, when you have more complicated correlation (for example, a correlation function like $\langle \sigma_0^x \sigma_3^x \sigma_5^x \sigma_6^x \rangle$) function the resulting matrices cannot be properly determined via Toeplitz matrices, which is a shame because constructing them is very quick.

Instead, we can construct the matrix by iterating over 2 point functions.

2.2 Explicit Example

Consider the correlation function

$$\langle \sigma_0^x \sigma_2^x \sigma_5^x \sigma_6^x \rangle = \langle \mathbf{B}_0 A_1 B_1 \mathbf{A}_2 \mathbf{B}_5 \mathbf{A}_6 \rangle$$

I have bolded the "endpoints" of strings to emphasize the form. In a gaussian free theory like the TFIM correlation functions like above can be solved via wick contractions. The Pfaffian of the matrix formed via elements of the wick contraction is an efficient way of doing this.

$$\langle \sigma_0^x \sigma_2^x \sigma_5^x \sigma_6^x \rangle = \det \begin{pmatrix} \langle B_0 A_1 \rangle & \langle B_0 A_2 \rangle & \langle B_0 A_6 \rangle \\ \langle A_1 B_1 \rangle & \langle B_1 A_2 \rangle & \langle B_1 A_6 \rangle \\ \langle A_1 B_5 \rangle & \langle A_2 B_5 \rangle & \langle B_5 A_6 \rangle \end{pmatrix} \quad (29)$$

We can see that because all quantities like $\langle A_i A_j \rangle = \delta_{ij}$ we can simply ignore these terms.

3 Kibble-Zurek Effect in the TFIM

In momentum space, the TFIM is 2x2 block diagonalizable into distinct bogoliouv modes with momentum k . That is to say, there are essentially N independent wavefunctions of the form

$$|k\rangle = a|0_k\rangle + b|1_k\rangle \quad (30)$$

The energy, Δ_k , between these two states (in the thermodynamic limit) is $2J|1 - g \cos(k)|$.

We can attempt to vary $H(g)$, the hamiltonian of the TFIM where g controls the strength of the $-\sigma_i^z$ coupling. When $g > 1$, the system is in the paramagnetic and when $g < 1$ it is in the ferromagnetic state.

If we vary $H(g)$ slower than a time scale set by $\tau_k = \frac{1}{\Delta_k}$ excitations of our initial state $|0_k\rangle$ are unlikely to occur. However, as $g = 1$ then τ_k diverges for low momentum modes and excitations form with probabilities close to unity. The system is said to "freeze out" as the time scale that the hamiltonian changes is far greater than the time scale needed for the system to equilibrate to the ground state.

When dynamics restart, the state—which did not follow the ground state through the transition—is now an excited state of a new hamiltonian and particles (essentially flipped spins) are released which are the defects of the kibble zurek mechanism. For instance, in the transition from the ferromagnetic state to the paramagnetic state, one might find $\langle \Psi(t) | \hat{\sigma}_i^z | \Psi(t) \rangle < 1$.

When the dynamics restart, the timescale needed for relaxation τ_k decreases and the system remains in the excited state with the same number of quasi-particles with exceedingly high probability. For this reason, we typically allow ourselves the following assumptions:

1: Far from the critical point we may extend our initial conditions to $t = -\infty$ and our final condition to $t = \infty$. 2: That all of the interesting physics only really happens in a small window of time near $t = 0$ where transitions like $|0_k\rangle \rightarrow |1_k\rangle$ occur with probability p_k .

The probability p_k of excitation for each mode k can be computed exactly in the thermodynamic limit and is found to have a Gaussian form in k for slow quenches. Specifically, for large τ_Q , the excitation probability is

$$p_k \approx \exp(-2\pi\tau_Q J k^2) \quad (31)$$

We will not derive this expression here, however we do acknowledge that this approximation is valid when $\frac{1}{\sqrt{4\pi J \tau_Q}} \ll \frac{\pi}{4}$.

The important (and measurable) thing is that now the expectation values of our operators should be different as the individual momentum modes are now in a statistical mixture. E.g. $|k(\infty)\rangle = |1\rangle$ with probability p_k , or more specifically

$$\rho_k = p_k |0\rangle \langle 0| + (1 - p_k) |1\rangle \langle 1| = \begin{pmatrix} p_k & 0 \\ 0 & 1 - p_k \end{pmatrix}$$

This is **not** a quantum superposition, so expectation values will not depend on overlaps such as $\langle 1_k | \hat{O}_k | 0_k \rangle$.

Now with those preliminaries, we can move on to the main calculation.

First, it will prove useful to evaluate the expectation values of combinations of creation and annihilation fermionic operators, e.g. calculating $\langle \psi | \hat{c}_k \hat{c}_q | \psi \rangle$, $\langle \psi | \hat{c}_k^\dagger \hat{c}_q | \psi \rangle \dots$ for $|\psi\rangle = |0_k\rangle, |1_k\rangle$.

Recall that this calculation is done by defining the fermionic operator $\hat{c}_k$ in terms of the bogoliouv operators (γ_k) that diagonalize the hamiltonian.

$$\hat{c}_k = u_k \gamma_k + v_{-k}^* \gamma_{-k}^\dagger \quad (32)$$

Recall that these operators are defined such that $\gamma_k |0\rangle = 0$ and $\gamma_k^\dagger |1\rangle = 0$.

3.1 Correlation functions in the TFIM

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We will do one explicit example of a correlation and cite the remaining results.

$$\langle 0 | \hat{c}_k \hat{c}_q | 0 \rangle = \langle 0 | (u_k \gamma_k + v_{-k}^* \gamma_{-k}^\dagger) (u_q \gamma_q + v_{-q}^* \gamma_{-q}^\dagger) | 0 \rangle \quad (33)$$

$$= \langle 0 | u_k \gamma_k v_{-q}^* \gamma_{-q}^\dagger | 0 \rangle \quad (34)$$

$$= \delta_{k,-q} u_k v_{-q}^* \quad (35)$$

We now cite all four combinations of correlation functions:

$$\langle 0 | \hat{c}_k \hat{c}_q | 0 \rangle = \delta_{k,-q} u_k v_{-q}^* \quad (36)$$

$$\langle 0 | \hat{c}_k^\dagger \hat{c}_q | 0 \rangle = \delta_{k,q} v_{-k} v_{-q}^* = |v_k|^2 \quad (37)$$

$$\langle 0 | \hat{c}_k \hat{c}_q^\dagger | 0 \rangle = \delta_{k,q} u_k u_q^* \quad (38)$$

$$\langle 0 | \hat{c}_k^\dagger \hat{c}_q^\dagger | 0 \rangle = \delta_{k,-q} v_{-k} v_{-q}^* \quad (39)$$

Now we will do the same with correlations involving the $|1_k\rangle = \gamma_k^\dagger |0\rangle$ state:

$$\langle 1_p | \hat{c}_k \hat{c}_q | 1_p \rangle = \langle 0 | \gamma_p \hat{c}_k \hat{c}_q \gamma_p^\dagger | 0 \rangle \quad (40)$$

$$= \langle 0 | \gamma_p (u_k \gamma_k + v_{-k}^* \gamma_{-k}^\dagger) (u_q \gamma_q + v_{-q}^* \gamma_{-q}^\dagger) \gamma_p^\dagger | 0 \rangle \quad (41)$$

$$= \langle 0 | (u_k \gamma_p \gamma_k + v_{-k}^* \gamma_p \gamma_{-k}^\dagger) (u_q \gamma_q \gamma_p^\dagger + v_{-q}^* \gamma_{-q}^\dagger \gamma_p^\dagger) | 0 \rangle \quad (42)$$

$$= \langle 0 | u_k u_q \gamma_p \gamma_k \gamma_q \gamma_p^\dagger + v_{-k}^* u_q \gamma_p \gamma_{-k}^\dagger \gamma_q \gamma_p^\dagger + u_k v_{-q}^* \gamma_p \gamma_k \gamma_{-q}^\dagger \gamma_p^\dagger + v_{-k}^* v_{-q}^* \gamma_p \gamma_{-k}^\dagger \gamma_{-q}^\dagger \gamma_p^\dagger | 0 \rangle \quad (43)$$

$$= \langle 0 | v_{-k}^* u_q \gamma_p \gamma_{-k}^\dagger \gamma_q \gamma_p^\dagger + u_k v_{-q}^* \gamma_p \gamma_k \gamma_{-q}^\dagger \gamma_p^\dagger | 0 \rangle \quad (44)$$

The terms with uneven amounts of creation and annihilation operators trivially get eliminated. We can evaluate the remaining terms using wick's theorem

$$\langle 0 | \gamma_p \gamma_{-k}^\dagger \gamma_q \gamma_p^\dagger | 0 \rangle = \delta_{p,-k} \delta_{q,p} + \delta_{-k,q} \quad (45)$$

$$\langle 0 | \gamma_p \gamma_k \gamma_{-q}^\dagger \gamma_p^\dagger | 0 \rangle = \delta_{k,-q} - \delta_{p,-q} \delta_{p,k} \quad (46)$$

so that

$$\langle 1_p | \hat{c}_k \hat{c}_q | 1_p \rangle = v_{-k}^* u_q (\delta_{p,-k} \delta_{q,p} + \delta_{-k,q}) + u_k v_{-q}^* (\delta_{k,-q} - \delta_{p,-q} \delta_{p,k}) \quad (47)$$